

Atlas of halving: what one refinement step does, and what emerges, in each dimension

Purpose, 2026-09-27 (user direction: a collective understanding of the halving of the lattice and what emerges in each case and dimension, as the mid-term goal). This note is the shared map. Each cell says what one halving does, what it leaves behind, and what survives the limit, with the status of the statement and the note that holds it. It adds no new theorem; it fixes one vocabulary so that the agents working here (Claude, GPT-6 Astra, Fable reviewers) and the user fill the same cells. Open cells are listed in § 5.

Status labels: **proved** (theorem in a repository note, reviewed or not as marked), **exact** (closed-form identity), **formal** (derivation assuming an unproved estimate), **known** (established literature, cited), **open**.

1. The operation

Halving one direction $a_1 \mapsto a_1/2$ of a lattice with plaquette heat times $t_{\mu\nu} = \lambda_D a_\mu a_\nu / \prod_{\rho \neq \mu, \nu} a_\rho$ (series/parallel note, eq. (1), small-field regime) splits exactly into two moves.

- **Series move.** A face containing the refined direction is cut in two; its heat time halves. Integrating the cutting edge is a convolution of the two half-weights: character coefficients multiply. For heat-kernel weights it closes exactly in every dimension.
- **Parallel move.** A face transverse to the refined direction gets a new copy in the mid-plane, both at doubled heat time. Integrating the mid-plane leaves the factor Ψ : a $(D-1)$ -dimensional gauge theory whose edges carry Brownian-bridge laws fixed by the coarse field. Pointwise products of weights add their logarithms; the heat kernel is closed under this only up to vortex terms (zero-spacing note, Proposition 1).
- **Count.** A halving produces parallel moves in $\binom{D-1}{2}$ transverse planes. An isotropic step is D directional halvings; the isotropic heat time scales as $t \propto \lambda_D a^{4-D}$.
- **Time-only halving** (Euclidean time at fixed spatial lattice) sends the transverse heat times to infinity; the Hamiltonian limit needs exponential (Wilson-type) magnetic weights (zero-spacing note, § 1).

Newton's halving is the one-dimensional case: inserting a time $t_{2.6}$ in a cell is a series move (refinement note, § 2). A physical record at the inserted time is a new variable attached to the body, the analogue of a parallel insertion. Forgetting a Gaussian record is an exact parallel move with defect proportional to $\hbar^2$, the identity for $\hbar = 0$ (record as parallel move).

1b. Cuts at any position (user remark, 2026-09-27)

Halving is the easiest cut; any cut at fraction $s \in (0, 1)$ of a cell performs the same two moves, with s as a parameter.

- **Series move.** The cut face splits into heat times st and $(1-s)t$; the convolution returns t exactly, for heat-kernel weights, in every dimension.

- **Parallel move.** In $D = 3$, with each transverse face weighted by its dual length (trapezoid rule), every old and new transverse face keeps heat time $2t$, as in halving. The inserted edges carry the bridge law at fraction s , variance $s(1-s)t_e$ in place of $t_e/4$. The free-field defect is (5) with $v \rightarrow v_s = 4s(1-s)v$ and coefficient $s(1-s)/(2(2t+v_s))$ on the layer differences in place of $1/(8(2t+v))$, so the size bound (6) scales exactly by $4s(1-s)$ (Corollary 2_s): halving is the largest single step, and an off-centre cut is gentler and shrinks the mesh less. Theorem 5 holds for every s (Corollary 5_s there).
- **Newton.** The cell action $K_\tau = F^2\tau^3/(24m)$ (two-path note) obeys $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, so one cut removes exactly $3s(1-s)K_\tau$, at most $\frac{3}{4}K_\tau$ (halving). Along any sequence of cuts whose mesh tends to zero the removed amounts sum to K_τ , whatever the positions and order (the remainder is $\sum_i C\tau_i^3 \leq C\tau \max_i \tau_i^2$). Proved, with the Cameron–Martin reading (each share is the energy of the Schauder hat the cut inserts), in the cut-measure note. The Galileo action is therefore an additive measure on the cut process: the refinement limit is cut-independent, and every cut spends a definite share of the action that Newton sends to zero. This is the Lévy–Ciesielski structure of the Brownian bridge (variance $s(1-s)$ at each dyadic or non-dyadic insertion) seen in the action.
- **Relative windings** (compact groups; 2026-09-28, refereed). On $U(1)$ the bridge midpoint at fraction s is a mixture over the total winding $W \in \mathbb{Z}$, with s -independent weights $\propto e^{-(\phi_e + 2\pi W)^2/(2t_e)}$, of wrapped Gaussians of variance $s(1-s)t_e$ centred at the interpolation shifted by $2\pi(1-s)W$; Theorem 5 holds for every s (Corollary 5_s). Halving is the only cut whose shifts form $\mathbb{Z}_2$, the parity mixture of Prop. 4; $s = p/q$ in lowest terms gives $\mathbb{Z}_q$, an irrational s a dense set. For a simply connected group the heat kernel's image sum runs over the coroot lattice $Q^\vee$, and the image midpoints sit at $e^{isX}e^{2\pi isH}$, $H \in Q^\vee$ (semiclassical placement; the exact weights carry Jacobians). These shifts are all central iff $sQ^\vee \subset P^\vee$, the coweight lattice ($P^\vee/Q^\vee$ is the centre). For $SU(2)$, $\frac{1}{2}Q^\vee = P^\vee$: at a halving a relative winding n acts by the centre element $(-1)^n$. Geometrically, $SU(2) = S^3$ and -1 is the antipode of e , so every closed geodesic through e passes through -1 at half length. For $SU(N)$, $N \geq 3$, only integer s satisfies the condition ($H = (1, -1, 0, \dots)$ forces $s \in \mathbb{Z}$); the shortest closed geodesics pass at half length through $\text{diag}(\dots, -1, \dots, -1, \dots)$, which is not central, and at $s = 1/N$ only $H \in NP^\vee$ give central shifts. Flipping one mid-edge by -1 flips every plaquette containing it, the smallest thin $\mathbb{Z}_2$ vortex in the sense of Mack and Petkova (1979); centre vortices go back to 't Hooft (1978), and factor-2 decimation with the $\mathbb{Z}_2$ factor kept explicit to Tomboulis (1981) (metadata for all three). Whether the bridge parity at a halving ties refinement to centre vortices is a question; the observation itself is elementary.
- **Which part of the centre a cut reaches** (2026-09-28, from the user's question on Dirac's belt trick; now Corollary 3 of the exact midpoint formula of the centre note, Astra referee pending). Lemma: for $SU(N)$ and a cut at $s = p/q$ in lowest terms, the relative windings land on exactly the subgroup $\mathbb{Z}_{\gcd(q,N)}$ of the centre $\mathbb{Z}_N$. Proof: a winding $H \in Q^\vee$ shifts the midpoint by $e^{2\pi ipH/q}$, central iff $pH/q \in P^\vee$, iff $H/q \in P^\vee$ (Bézout, since $H \in Q^\vee \subset P^\vee$); so the central shifts are $(P^\vee \cap q^{-1}Q^\vee)/Q^\vee$, the q -torsion of $P^\vee/Q^\vee \cong \mathbb{Z}_N$. $\square$ Dyadic refinement therefore reaches the 2-primary part of the centre: all of it for $SU(2)$ at the first halving, nothing for $SU(3)$ at any depth, while one trisection reaches all of $\mathbb{Z}_3$. Centre elements are focal points: the shortest geodesics from e to -1 in $SU(2)$ form an S^2 , those from e to $\omega \cdot 1$ in $SU(3)$ a $\mathbb{CP}^2$, and the closed geodesic along $(1, 1, -2)$ passes $\omega \cdot 1$ and

$\omega^2 \cdot 1$ at its thirds. For $SU(2)$ this is Dirac's belt trick, $\pi_1(SO(3)) = \mathbb{Z}_2$ (Newman 1942, metadata): the halving turns w extra 720° turns of a bridge into w extra 360° turns at its midpoint, and the exact formula shows it, since in Theorem 3 of the $SU(2)$ midpoint note the image sums enter with $(-1)^w$ for half-integer J (Ξ_-/Ξ_+) and drop out of the ratio for integer J . In $1+3$ the internal and the spatial $\mathbb{Z}_2$ meet in “spin from isospin” (Jackiw and Rebbi 1976; Hasenfratz and 't Hooft 1976; metadata). For $SU(3)$ the phase a centre-landing winding leaves on a character is its triality, so quark representations see ω and gluons do not; the exact trisection formula is (1) of the centre note at $s = \frac{1}{3}$. Per step, the smallest centre-landing trisection winding costs $e^{-24\pi^2/t}$ against $e^{-8\pi^2/t}$ for root windings ($|H|^2 = 2 \text{tr} H^2 = 12$ and 4), both beyond the $1+3$ threshold $c > 2/b_0$. The lemma concerns which classes the windings of a step can reach; centre-valued link configurations exist on any lattice, and thick vortices are a separate question.

- **The rod** (user remark, 2026-09-27). The dialecticians' stick of Zhuangzi 33, 一尺之捶，日取其半，萬世不竭 (“a one-foot stick, each day take half, in ten thousand generations it is not exhausted” ; text, attributed to the 辯者, the debaters, and not to Hui Shi personally), is Zeno's dichotomy as a cut schedule: every day one cut, at $s = \frac{1}{2}$ of the remaining piece only. The mesh stays at $\tau/2$, so this schedule is no refinement, and the action identity above prices it exactly: day k removes $\frac{3}{4}K_\tau 8^{-(k-1)}$, the total spent is $\frac{3}{4} \cdot \frac{8}{7}K_\tau = \frac{6}{7}K_\tau$, and $\frac{1}{7}K_\tau$ stays in the cut-off pieces. The stick is inexhaustible and so is the action: $\frac{1}{7}$ of it is never reached by this schedule. The remainder converges to a point, and the Mohist Canon B names that point: 非半弗斷，則不動，說在端 (“without halving there is no cutting and no moving; the explanation lies in the 端, the point”), with the explanation 前則中無爲半，猶端也。前後取，則端中也 (text; working gloss: “taking from the front, the middle is never halved, and one ends as at a point; taking from front and back, the point is in the middle”). The two Mohist protocols, one-ended and two-ended, both converge to a point and neither refines the whole stick. Refinement, which spends all of K_τ , needs every piece cut again. Resemblance with Zeno is recorded as convergence; transmission to China would need evidence this note does not have.
- **The frame.** A cut at an arbitrary, even irrational, position is the geometers' cut of Book I's closing scholium. The questions for each cell become: does the limit depend on the cut sequence (for Newton, no, proved in the cut-measure note; for the free field, expected), and does the universal part (the $\log 2$ of cell 4, which for general cuts should read $\log(1/s)$ summed over the step) survive when s varies (conjecture: yes, as the scheme-independent one-loop coefficient).

2. The atlas by dimension

$1 + n$	Series move	Parallel planes per halving	What one step leaves	What survives the limit	Status and notes
$0 + 0$	none (a single weight)	none	the weight $k_t(U)$ itself; a single integral with an $e^{-c/\hbar}$ structure	a large- N transition (Gross–Witten)	known; dimension ladder
$1 + 0$ Newton	exact after one cubic counterterm $-F^2\hbar^3/(24M)$; the defect $-F^2uv\hbar/(8M)$ is a coboundary	none	a scalar	unobserved: the exact propagator at every $\hbar$; recorded: a floor only if joint determinacy fails	proved; refinement note, fifth postulate
$1 + 1$	exact (heat-kernel convolution)	0	nothing for heat kernels; for any action, $a^{-2}(1 - \hat{c}_R)$	a conjugation-invariant Lévy exponent $\psi(R)$; Yang–Mills iff Lindeberg; circle spectrum $\hbar cL\psi(R)$; no local particle	proved (Theorem 2 of the zero-spacing note); known (Lévy)
$1 + 2$	exact	1	free: an irrelevant quadratic form of relative size $a^2K^2/16 + (Ga)^2/8$; $U(1)$: vortices of density $e^{-\pi^2/(8t)}$; $SU(N)$: coupling shifts $O(t)$, curvature term explicit	free photon for $U(1)$ at fixed λ_3 (every lattice gap dies); conjectured gap $C_3\hbar c\lambda_3$ for $SU(N)$	proved for free and $U(1)$; formal for $SU(N)$ (Proposition 7); see § 3

$1 + n$	Series move	Parallel planes per halving	What one step leaves	What survives the limit	Status and notes
$1 + 3$	exact	3	$t = g^2$ fixed: coupling shifts that recur each step; large fields suppressed only as $(a\Lambda)^{2b_0c}$	running coupling, $\Lambda = \mu e^{-1/(2b_0g^2)}$; small instantons die $(11N/3 > 4)$; dislocations need $c > 2/b_0$ ($\frac{6}{11}$ of an instanton for $SU(2)$); $U(1)$: Coulomb phase	known (running, 6/11 criterion, Driver); the measure-level route fails for $U(1)$ (proved)
$1 + n, n \geq 4$	exact	$\binom{n}{2}$	t grows as $a \rightarrow 0$	no small-field regime; a phase transition, no established continuum limit	known (Creutz for $SU(2)$ in 5D)

The same heat time $t = \hbar g_{\text{cl}}^2 a^{4-D}$ controls the column “what one step leaves” . For $D < 4$ refining and taking $\hbar \rightarrow 0$ push it the same way; in mechanics refining makes each cell more quantum (dimension ladder, § 2).

3. The atlas for $1 + 2$ by group

Group	Isolated cube	Full mid-plane	One step, measure level	Iteration and limit
free ($\mathbb{R}^n$)	exact	exact defect (Proposition 2 of the series/parallel note)	Gaussian, exact	blocked actions converge to a Gaussian fixed point (known, Bell–Wilson; for this blocking asserted)

Group	Isolated cube	Full mid-plane	One step, measure level	Iteration and limit
$U(1)$	exact (Proposition 3)	exact charge form, vortex bound (Proposition 4, Corollary)	one step equals the free step up to density $e^{-\pi^2/(8t)}$ (Theorem 5, refereed); whole measure within TV $2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}$ of the monopole-free part (monopole note)	free photon (Gross 1983, known)
$SU(2)$	midpoint characters exact for every spin; spin- $\frac{1}{2}$ and spin-1 cube sectors exact, with matrix image bounds for spin 1 (closed form, Theorems 1–4)	formal order- t expansion: cut shifts positive, transverse negative; other local operators have dimension at least six; explicit torus winding term	formal; needs normalized estimate (18) with winding control	open
$SU(3)$	curvature term explicit, softening $t_j X_j ^2/512$ (group-general form of Proposition 6)	open	formal	open; the gap $C_3 \hbar c \lambda_3$ is the infrared obligation

4. What emerges, read across the atlas

- **Series moves never produce anything new.** In every dimension they close exactly for heat-kernel weights, and in mechanics after one scalar counterterm. The one-dimensional Newton problem and the two-dimensional gauge theory are all series.
- **Parallel moves carry everything that renormalizes.** Their number, $\binom{D-1}{2}$, is zero exactly where the theory is exactly soluble.
- **Additivity is universal; locality is one-dimensional** (2026-09-28). Integrating out the fine variables of a Gaussian step leaves the Schur complement of the action, its minimum over the fine variables with the coarse ones fixed (the Dirichlet principle), and successive Schur complements compose exactly ((13) of the four-dimensional note). In Newton's cell this additivity is Theorem 2 of the cut-measure note: the hats of successive cuts are orthogonal and their energies add

up to K_τ . What one dimension alone grants is locality: the minimizer between two cut points is the chord, so the blocked action is again nearest-neighbour and the series move closes; $1 + 1$ gauge theory, with no transverse planes, shares it. With transverse planes the minimizer is coupled across the mid-space, the Schur complement is non-local, and its local truncation error is the parallel defect: (5) and Corollary 2_s of the series/parallel note split it into a displacement part, which at $v = 0$ is the convexity gap of the interpolated flux, and a noise part from relaxing the interpolant.

- **The exponential ladder.** Each compact correction left by a step is $e^{-c/t}$ (vortices, monopoles, dislocations). In $D < 4$ it is summable per physical volume at fixed coupling, so compact effects die; in $D = 4$ the running turns it into a power of a , so survival is a threshold condition; in $D > 4$ there is no small t . In $1 + 3$ at one loop, $a\Lambda = e^{-1/(2b_0t)}$ with $t = \hbar g^2(a)$, so $e^{-c/t} = (a\Lambda)^{2b_0c}$ and $2b_0c$ is the scaling dimension of the correction. The generated gap is itself on the ladder: $am \propto e^{-1/(2b_0\hbar g^2)}$ is the member with $2b_0c = 1$. The bridge windings sit far above the per-volume threshold $2b_0c = 4$: $2b_0c = 11N/3$ for root images ($c = 8\pi^2$) and $11N$ for the $SU(3)$ centre images of a trisection ($c = 24\pi^2$, centre note) (2026-09-28).
- **What survives is trajectory-dependent.** Compact $U(1)$ in $1 + 2$ keeps a gap along $\lambda_3 a \simeq c_0/(2 \log(1/a))$, $c_0 \approx 4.99$ the monopole exponent, where the monopole density per physical volume diverges, and loses it at fixed λ_3 (Göpfert–Mack versus Gross).
- **The action floor survives what removes the mass gap** (user observation, 2026-09-28). The heat time is $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$, so the compact corrections a step leaves, $e^{-c/t} = e^{-c/(\hbar g_{\text{cl}}^2 a)}$, vanish as $a \rightarrow 0$ at fixed coupling and as $\hbar \rightarrow 0$ at fixed g_{cl} and a , while the $w = 0$ Gaussian term of (9) of the series/parallel note, whose width is set by $\hbar$, is what remains. Both $U(1)$ continuum limits on record are Gaussian: Gross’ s photon, with field-strength covariance $\lambda_3 K_0 = \hbar g_{\text{cl}}^2 K_0$ at fixed g_{cl} , and Göpfert–Mack’ s free massive scalar, canonically normalized along a trajectory on which $\lambda_3 = \hbar g_{\text{cl}}^2$ diverges. The refinement at fixed coupling removes the lattice mass gap and keeps the unit of action. And $\hbar$ was first measured in this same gapless theory in $1 + 3$, cavity radiation (Planck’ s talk of 14 December 1900, written up as Ann. Phys. 309, 553 (1901), through the resonators’ energy elements; the field-mode reading is Debye 1910, metadata); in a finite cavity the supplied box gap $2\pi\hbar c/L$ remains, and “gapless” is the infinite-volume statement. The atlas thus carries two gaps of different kinds: a mass gap, generated and trajectory-dependent, and an action floor, a unit that every cell of every limit keeps (each cut weighted by $e^{-\text{share}/\hbar}$, cut-measure note, Prop. 4). In the terms of G07 both are unit multiples of the fixed constants (its Theorem 1); the action floor is supplied, $\hbar$ written into $e^{-S/\hbar}$, while a generated gap needs two scale-free structures that do not commute (G07 § 1; G08).
- **Constants that emerge.** Mechanics: one scalar counterterm per cell, and one action constant if joint determinacy is denied (Theorems B and B’ of the fifth-postulate note). $1 + 1$: the Lévy exponent (one coupling for Yang–Mills). $1 + 2$: one coupling, shifted summably. $1 + 3$: one scale Λ by dimensional transmutation.

5. Open cells

1. $SU(2)$ cube: filled through $J = 1$ (Theorem 4, exact diagonal entries, image bounds and cube sector); $J > 1$ entries and control of the full spin sum remain open.
2. $SU(2)$ full mid-plane in $1 + 2$: formal order- t calculation filled, refereed (all items

- accepted): cut shifts positive, transverse negative, remaining bulk terms of dimension six; the fixed-torus winding term is removed by the clause $\min_j N_j \geq c_0/t$; open: the uniform estimate (18).
3. Stability of Hypothesis P(α) under iteration. For $U(1)$ in 1 + 2, reduced to exact Gaussian blocking (Corollary 2' of the monopole note); open for $SU(N)$.
 4. 1 + 3: formal anisotropic one-step integrals and conditional logarithm; the massive mid-space sums are infrared regular. Open: composition with generated interactions and finite endpoint matching; equal endpoint schemes give $-2b_0 \log 2$ analytically.
 5. Time-only halving. Structural reading, 2026-09-27, of known results. At fixed spatial lattice the electric faces make the series moves: heat-kernel electric weights are a semigroup in the time step, so they compose exactly. The magnetic faces make the parallel moves: with exponential weights $e^{-a_0 V}$ they close exactly in their own family, since products of such weights add the exponents. The whole defect of a time halving is the non-commutation of the two families, and it vanishes as $a_0 \rightarrow 0$ by the Trotter product formula (Trotter 1959; Chernoff 1968; metadata; strong convergence for the Laplacian on G^E plus a bounded magnetic potential). This is why the Hamiltonian limit (Kogut and Susskind 1975; transfer matrix: Creutz 1977; metadata) exists at every fixed spatial lattice, while spatial halving leaves the non-closing factor Ψ . Open: an operator-norm rate, which needs domain estimates for the commutator of the Laplacian with V .
 6. Filled 2026-09-27 for Gaussian records (record as parallel move): forgetting a record convolves momentum with variance $\hbar^2/(4\sigma^2)$; precisions add in parallel; refinement converges iff $\sum \sigma_j^{-2} < \infty$; identity for $\hbar = 0$.
 7. The infrared: from ultraviolet control in 1 + 2 to $C_3 > 0$.

6. Consequence for STATE

STATE carries the mid-term goal and points here. Each result that fills a cell updates the cell in this note and nothing else; the notes that prove results remain the record.